

Second and Higher Derivatives: Motion, Decay, and Implicit Curves

- Every model below states its units. Keep them attached to your work.
- Write derivatives in the notation the problem uses ($s''(t)$, $y^{(4)}$, $\frac{d^2y}{dx^2}$).
- Leave exact values (fractions, radicals, e) unless a decimal is asked for.

1. A freight elevator's height above the ground floor is modelled by $h(t) = 2t^4 - 5t^3 + 7t$, where h is in metres and t is in seconds. Find the elevator's acceleration function $h''(t)$.

Answer: _____

2. A test cart runs along a straight track with position $s(t) = t^3 - 9t^2 + 24t$, where s is in metres and t is in seconds.

- (a) Find the acceleration function $s''(t)$.

Answer: _____

- (b) Find the cart's acceleration at $t = 5$ seconds.

Answer: _____ m/s²

3. A buoy bobs on a swell so that its height above mean sea level is $y(t) = 3 \sin(2t)$ metres, with t in seconds. Find the fourth derivative $y^{(4)}(t)$.

Answer: _____

4. A cooling tank loses water so that the volume remaining is $V(t) = 500e^{-t/5}$ litres, with t in minutes. Find the third derivative $V'''(t)$.

Answer: _____

5. A survey drone's altitude is $A(t) = t^4 - 8t^3 + 18t^2 + 30$ metres, with t in seconds and $0 \leq t \leq 6$.

(a) Find the vertical acceleration $A''(t)$.

Answer: _____

(b) Find every time t in the interval at which the vertical acceleration is zero.

Answer: _____

6. A hanging cable is modelled near its lowest point by the curve $y^2 = 4x$ with $y > 0$, where x and y are in metres.

(a) Differentiate implicitly to find $\frac{dy}{dx}$ as a function of x alone.

Answer: _____

(b) Find $\frac{d^2y}{dx^2}$ as a function of x alone.

Answer: _____

7. A circular fairground ride has rim $x^2 + y^2 = 25$, with x and y in metres. A camera tracks the upper half of the rim, where $y > 0$.

- (a) Find $\frac{d^2y}{dx^2}$ for the upper half, as a function of x alone.

Answer: _____

- (b) Evaluate $\frac{d^2y}{dx^2}$ at the point $(3, 4)$. Give an exact fraction.

Answer: _____

8. A bacterial culture holds $P(t) = t^3 - 12t^2 + 36t + 50$ thousand cells after t hours, for $0 \leq t \leq 8$.

(a) Find $P''(t)$.

Answer: _____

(b) The growth rate is $P'(t)$. Find the hour at which the growth rate is smallest.

Answer: _____

(c) Explain how the sign of P'' shows that the growth rate really is smallest there, and not largest.

Answer: _____